`myVariable`, `mYvAlUe`, `DELAY_TIME`, and `value1` are all valid names. Conversely, the variable names `1stValue`, `delay time`, and `time&money` are not valid names for a variable.

Now some of these names may not be the most readable, which is why many Arduino programmers have adopted certain naming guidelines or **conventions**. Naming conventions are generally controversial in programming circles, and they are not specifically required when defining your own unique names, but they can make code easier to read in the big scheme of things.

For example, we can have a variable name with multiple words but, because we cannot use spaces inside our variable name, we might want a way to separate each word to make the name easier to read. One possibility is to use an underscore between each word, as in `red_led_connected_to_pin_9`, a method usually favored by UNIX programmers. Instead, we will use the convention known as **camelback notation**, as in `redLEDConnectedToPin9`. Under this convention, the first letter of the first word will be lowercase while the first letter of each subsequent word will be capitalized with no space between each word. Take the following examples:

```
variable

myVariable

value1

tiltSwitchValue
```

It's really your choice how you name your variables and you will often see both types in other people's sketches, but we feel that camelback notation makes the code easier to read and matches the format the Arduino developers have decided upon for all of the built-in functions in the Arduino library. Likewise, it is the general convention that constants are sometimes written in all caps with underscores in between multiple words, as in the following examples, although we will not adhere strictly to this in naming our own constant variables.

```
HIGH

INPUT

BLUE_LED
```

■ **Note** Remember that the Arduino language is case-sensitive, so whatever style you adopt, you need to use it consistently throughout your sketch. This also means that any predefined variables or function names that are already a part of Arduino C must be used with exact capitalization.

One thing to keep in mind is that while there is no maximum or minimum length for variable names, extremely long names start to become rather tedious to type over and over. On the other hand, one-character names are often too cryptic. A few exceptions to this would be the lower case letter `i`, which is a commonly used variable to indicate an index along with the variable names `j` and `k`. Likewise, `a`, `b`, `c` or `x`, `y`, `z` are other sometimes used as short names, often when adding a longer name would not necessarily provide any more information.